

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC HOMEWORK & ASSESSMENT SUBMISSION

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Course Code & Title:	21MAB201T - TRANSFORMS AND BOUNDARY VALUE PROBLEMS
Task & Channel:	Fourier Series & Harmonic Analysis Assignment 2 [EMAIL]

Assignment Description:

Calculate the Fourier expansion for $f(x) = x^2$ in $(-\pi, \pi)$ and deduce Riemann Zeta(2) sum. Submit handwritten/typeset PDF.

Technical Solution:

Solution: Fourier Series of $f(x) = x^2$ in $(-\pi, \pi)$

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1. Fourier Expansion Formula:

Since $f(x) = x^2$ is an even function ($f(-x) = f(x)$), the sine coefficient $b_n = 0$.

The Fourier series is:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx)$$

2. Calculation of Coefficients:

- a_0 Evaluation:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{2\pi^2}{3}$$

Therefore, $\frac{a_0}{2} = \frac{\pi^2}{3}$.

- a_n Evaluation (Integration by parts):

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx = \frac{2}{\pi} \left[x^2 \frac{\sin(nx)}{n} + 2x \frac{\cos(nx)}{n^2} - 2 \frac{\sin(nx)}{n^3} \right]_0^{\pi}$$

$$a_n = \frac{2}{\pi} \left(\frac{2\pi}{n^2} (-1)^n \right) = \frac{4(-1)^n}{n^2}$$

3. Complete Series:

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx)$$

4. Deduction of Basel Sum:

Setting $x = \pi$ (point of continuity):

$$\pi^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$